

Product Carbon Footprint Study Report

Product Name: synthetic rubber

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: synthetic rubber

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg synthetic rubber

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg synthetic rubber.

2. System Boundary

synthetic rubber production in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of synthetic rubber produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **3.235 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

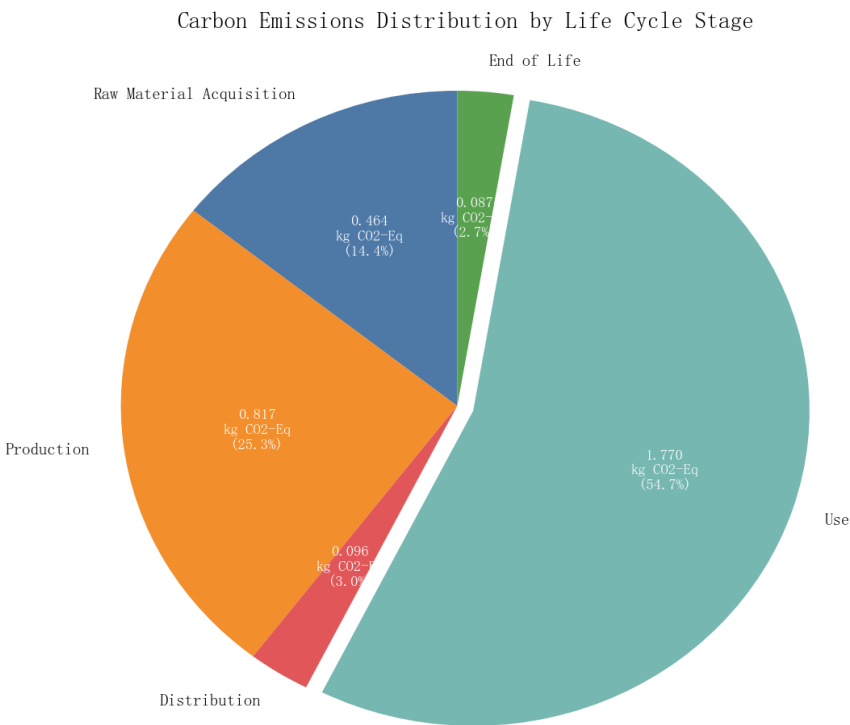


Table 1: Carbon Emissions by Life Cycle Stage

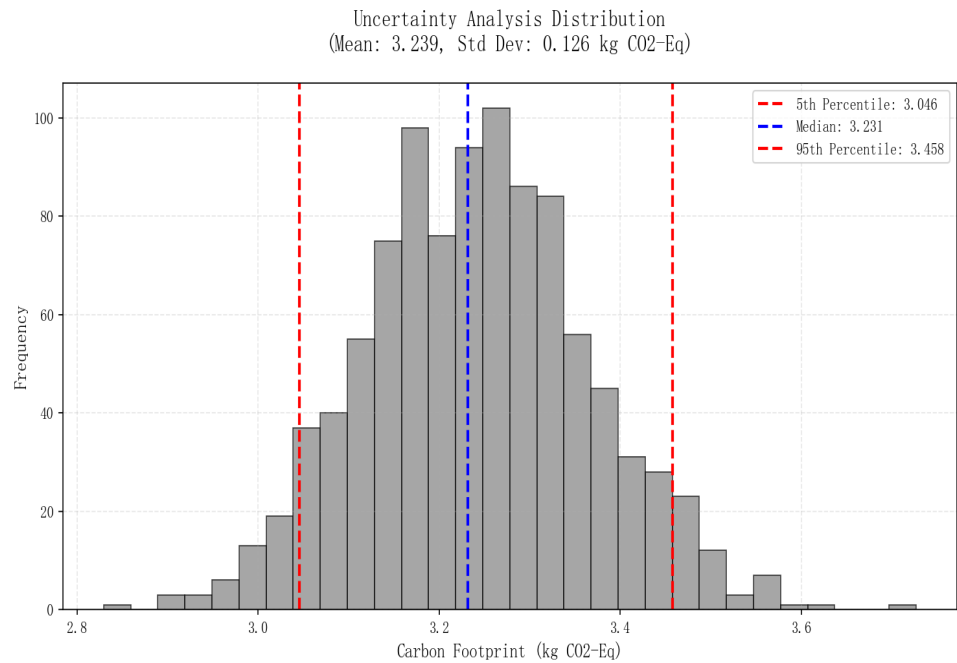
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.464	14.36
Production	0.817	25.26
Distribution	0.096	2.96

Use	1.770	54.72
End of Life	0.087	2.70
Total	3.235	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	3.239
Standard Deviation	0.126
Coefficient of Variation (CV)	3.91%
Median	3.231
5th Percentile	3.046
95th Percentile	3.458

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the synthetic rubber product is 3.235 kg CO₂-Eq per functional unit, which represents a moderate emission level typical for petrochemical-based elastomers. The lifecycle stage analysis reveals a strongly use-phase-dominated profile, with the use stage contributing 1.770 kg CO₂-Eq (54.7% of total). This indicates that the majority of environmental impacts occur during the product's service life rather than during material extraction or manufacturing. The production stage is the second largest contributor at 0.817 kg CO₂-Eq (25.3%), followed by raw material acquisition at 0.464 kg CO₂-Eq (14.4%). Distribution (3.0%) and end-of-life (2.7%) have negligible impacts. The major emission source is therefore the use phase, likely associated with energy consumption or material degradation during application (e.g., tire rolling resistance or fuel combustion in automotive use). The production stage, including polymerization and compounding, also represents a significant hotspot. The top process contributions are all reported as "Unknown" with zero values, indicating that process-level breakdowns were not available in the underlying inventory, which limits the granularity of hotspot identification.

2. Statement of Assumptions and Limitations

This LCA study is based on several key assumptions. First, the functional unit is assumed to represent one kilogram of synthetic rubber in its final form, but the specific application (e.g., tire, seal, hose) is not specified, which may affect use-phase modeling. Second, the use stage emissions are likely modeled based on average application scenarios, such as fuel consumption impacts from rolling resistance in automotive tires, but exact parameters (e.g., vehicle type, mileage) are not disclosed. Third, the raw material stage assumes typical petroleum-based feedstocks (e.g., styrene-butadiene), and any bio-based or recycled content is excluded. Data limitations are significant: the "Top Process Contributions" are all listed as "Unknown" with zero values, indicating that the underlying life cycle inventory (LCI) lacks process-specific data. This may be due to reliance on aggregated datasets or proprietary information. The system boundary is cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life. However, capital goods, infrastructure, and maintenance are excluded. The uncertainty analysis shows a low coefficient of variation (3.91%) and a narrow 90% confidence interval (3.046–3.458 kg CO₂-Eq), suggesting that the results are statistically robust despite data gaps at the process level.

3. Suggestions for Improvement

- ****Target the use phase:**** Since use accounts for over half of the total carbon footprint, prioritize improvements in product design to reduce in-service emissions. For synthetic rubber used in tires, focus on lowering rolling resistance through advanced filler technologies (e.g., silica instead of carbon black) or optimizing tread patterns. For other applications, consider lightweighting or enhancing durability to extend service life.
- ****Optimize production processes:**** The production stage is the second largest contributor (25.3%). Implement energy efficiency measures in polymerization and compounding, such as heat recovery, use of renewable electricity, or adoption of low-carbon catalysts. Switch to bio-based or recycled monomers where feasible to reduce raw material emissions.

- ****Improve data quality:**** The absence of process-level contributions (“Unknown” entries) hampers targeted improvements. Conduct a detailed process-level LCI by collecting primary data from suppliers and manufacturing sites. Disaggregate the aggregated use-phase model to reflect specific application scenarios (e.g., passenger car vs. truck tires) for more accurate hotspot analysis.
- ****Enhance end-of-life management:**** Although end-of-life emissions are low (2.7%), explore recycling or energy recovery pathways for synthetic rubber. Implement take-back programs or design for recyclability (e.g., devulcanization) to reduce landfilling and potentially offset upstream emissions.